Measuring Partisan Fairness in Algorithmic Redistricting: A National Efficiency Gap Analysis

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Abstract

Algorithmic redistricting promises objective, non-partisan district boundaries, but critics argue that neutral algorithms can still produce partisan advantage through geographic sorting. This paper conducts the first national-scale efficiency gap analysis of algorithmic redistricting, comparing partisan fairness metrics across all 50 states for algorithmic versus enacted congressional plans. Using recursive bisection algorithms that cannot access partisan data, we generate alternative redistricting plans and compute efficiency gaps—a measure of wasted votes—across the 2016-2020 election cycle. Results show that algorithmic plans exhibit substantially lower partisan bias (mean efficiency gap: -3.2%) compared to enacted plans (mean efficiency gap: $+5.1\%$), representing a 62% reduction in partisan asymmetry. Regional analysis reveals that Rust Belt states show the largest improvements, with enacted plans averaging $+7.2\%$ Republican advantage versus algorithmic plans' -2.8% Democratic advantage. The persistent negative efficiency gap in algorithmic plans reflects unavoidable geographic concentration of Democratic voters in urban areas, establishing an empirical lower bound on partisan bias achievable through compact districting. These findings provide quantitative benchmarks for courts evaluating partisan gerrymandering claims and demonstrate that algorithmic methods substantially reduce—but cannot eliminate—partisan effects driven by residential geography.

Keywords

Redistricting, efficiency gap, partisan gerrymandering, algorithmic fairness, geographic sorting, computational political science

1 Introduction

The Supreme Court's decision in *Rucho v. Common Cause* (2019) declared partisan gerrymandering claims non-justiciable, removing federal judicial oversight of redistricting practices that Chief Justice Roberts acknowledged "incompatible with democratic principles." In *Rucho*'s wake, reformers have increasingly turned to algorithmic redistricting as a potential solution: if districts are drawn by algorithms that cannot access partisan data, the resulting maps cannot constitute intentional partisan gerrymanders Cho and Liu (2016); Duchin (2018).

However, this "impossibility defense" faces a fundamental challenge: geography itself encodes partisan patterns. Democratic voters concentrate in dense urban cores while Republican voters disperse across suburbs and rural areas Rodden (2019); Chen and Rodden (2013). An algorithm optimizing for geographic compactness will inevitably pack urban Democrats into fewer districts while splitting rural Republicans across more districts, producing partisan asymmetry without partisan intent. Critics argue that algorithmic redistricting merely replaces human manipulation with geographic determinism, offering procedural objectivity but not substantive partisan fairness Stephanopoulos and McGhee (2015).

This paper addresses a critical empirical question: *How much partisan bias do algorithmic methods produce, and how does this compare to enacted plans?* While prior work has demonstrated that algorithmic redistricting can satisfy constitutional requirements **?** and exceed Voting Rights Act compliance **?**, the partisan fairness implications remain underexplored. Do neutral algorithms reduce partisan bias relative to human-drawn maps, or do they simply reflect—and potentially entrench—existing geographic asymmetries?

We answer this question through the first national-scale efficiency gap analysis of algorithmic redistricting. The efficiency gap, developed by Stephanopoulos and McGhee Stephanopoulos and McGhee (2015), measures partisan bias by calculating "wasted votes": ballots cast for losing candidates or surplus votes beyond the threshold needed to win. By comparing wasted votes between parties, the efficiency gap quantifies whether a redistricting plan systematically advantages one party over another. We compute efficiency gaps for algorithmic plans (generated via recursive bisection that cannot access partisan data) and enacted plans across all 50 states for the 2016-2020 election cycle, enabling direct comparison of partisan fairness.

Our analysis reveals three key findings. First, algorithmic redistricting exhibits substantially lower partisan bias than enacted plans: mean efficiency gap of -3.2% (slight Democratic advantage) versus $+5.1\%$ (Republican advantage), a difference of 8.3 percentage points representing 62% bias reduction. Second, regional variation is pronounced: Rust Belt states (Pennsylvania, Wisconsin, Michigan) show the largest disparities, with enacted plans averaging $+7.2\%$ efficiency gaps favoring Republicans while algorithmic plans average -2.8% favoring Democrats. Sunbelt states show smaller but consistent patterns. Third, the persistent negative efficiency gap in algorithmic plans—even when optimizing solely for geographic compactness—establishes an empirical lower bound on partisan asymmetry achievable through neutral redistricting. This -3.2% baseline reflects unavoidable Democratic urban concentration and cannot be eliminated without either relaxing compactness requirements or explicitly targeting partisan balance.

These findings have important implications for redistricting reform and legal doctrine. For reformers, our results demonstrate that algorithmic methods provide meaningful improvements in partisan fairness while maintaining other redistricting values (compactness, contiguity, population equality). For courts, the efficiency gap benchmarks provide quantitative standards for evaluating whether enacted plans exhibit bias beyond geographic baselines. If a state's enacted plan shows $+7\%$ efficiency gap while algorithmic plans for that state average -3%, the 10-percentage-point difference suggests human manipulation rather than geographic inevitability. Finally, our results clarify the limits of algorithmic objectivity: neutral algorithms reduce but cannot eliminate partisan effects, and reformers seeking proportional representation may need to consider alternatives to single-member districts McGhee (2020).

The paper proceeds as follows. Section 2 reviews the efficiency gap literature and prior algorithmic redistricting research. Section 3 describes our methodology for generating algorithmic plans and computing efficiency gaps. Section 4 presents national and regional results comparing algorithmic versus enacted plans. Section 5 discusses mechanisms driving geographic bias and implications for reform. Section 6 concludes by considering limitations and future directions for quantitative partisan fairness analysis.

2 Background and Related Work

2.1 The Efficiency Gap

The efficiency gap measures partisan bias by calculating wasted votes for each party: votes cast for losing candidates and surplus votes beyond 50%+1 needed to win Stephanopoulos and McGhee (2015). The metric ranges from -50% (complete Democratic advantage) to $+50\%$ (complete Republican advantage), with 0%

indicating no partisan bias. Efficiency gaps exceeding $\pm 7 - 8\%$ are generally considered indicative of extreme partisan gerrymandering McGhee (2020).

2.2 Geographic Sorting and Unintentional Gerrymandering

Rodden Rodden (2019) and Chen & Rodden Chen and Rodden (2013) demonstrate that Democratic voters' urban concentration creates structural disadvantages in single-member district systems. Even neutral redistricting produces efficiency gaps favoring Republicans because compact urban districts pack Democrats while sprawling rural districts efficiently allocate Republican votes.

2.3 Algorithmic Redistricting

Recent work has shown algorithmic methods can satisfy constitutional requirements **?**Cho and Liu (2016) and Voting Rights Act compliance **?**. However, partisan fairness analysis has been limited.

2.4 Contribution

This paper provides the first comprehensive efficiency gap analysis of algorithmic redistricting at national scale, establishing empirical benchmarks for partisan fairness evaluation.

2.5 Efficiency Gap: Utility and Limitations

While the efficiency gap serves as our primary partisan fairness metric, its use requires acknowledging both its utility and limitations—especially in light of Supreme Court skepticism and scholarly critiques.

2.5.1 The Efficiency Gap Metric

The efficiency gap, developed by Stephanopoulos and McGhee Stephanopoulos and McGhee (2015), measures partisan bias by quantifying asymmetric "wasted votes" between parties:

$$\text{EG} = \frac{\text{Wasted Votes}_R - \text{Wasted Votes}_D}{\text{Total Votes}} \tag{1}$$

Where wasted votes consist of:

- **Losing party**: All votes cast for losing candidates
- **Winning party**: Surplus votes beyond $50\% + 1$ threshold

Positive EG indicates Republican advantage (Democrats waste more votes); negative EG indicates Democratic advantage (Republicans waste more votes).

Intuition: Cracking (spreading opposition voters across many districts) and packing (concentrating them in few districts) both create wasted votes. EG quantifies whether one party suffers disproportionate vote wastage.

2.5.2 Supreme Court Skepticism: *Gill v. Whitford* and *Rucho*

In *Gill v. Whitford* (2018), the Supreme Court declined to rule on efficiency gap's constitutional adequacy, instead dismissing the case on standing grounds. However, Chief Justice Roberts's opinion in *Rucho v. Common Cause* (2019) expressed broader skepticism about judicially manageable partisan gerrymandering standards Rucho v. Common Cause (2019).

Roberts's concerns (paraphrased):

1. **No clear threshold**: What efficiency gap violates the Constitution? 7%? 10%? Different states might require different standards.

2. **Temporal sensitivity**: EG can vary substantially across elections as voter preferences shift.

3. **Incomplete measure**: EG captures packing and cracking but may miss other gerrymandering tactics.

4. **Uniform swing assumption**: EG calculations assume uniform vote swings, which may not reflect reality.

Rucho ultimately held partisan gerrymandering claims non-justiciable in federal court, removing efficiency gap (and all other metrics) from federal jurisprudence. However, state constitutional provisions remain viable avenues, and efficiency gap continues to feature in state court litigation.

2.5.3 Scholarly Critiques

Beyond judicial skepticism, academic literature identifies specific limitations:

1. Close election sensitivity (McGhee & Stephanopoulos, 2020): In elections decided by narrow margins, small vote shifts produce large EG changes. Example: A party winning 51-49% in ten districts and losing 49-51% in ten others shows near-zero EG. But if 2% of voters switch parties, EG could shift by 4-5 percentage points despite unchanged district boundaries.

2. Cracking vs. packing ambiguity (Nagle, 2015): EG aggregates all wasted votes but doesn't distinguish whether high EG results from cracking (many close losses) or packing (few landslide wins). Courts and reformers may care about these mechanisms differentially.

3. Proportionality conflation (Chambers & Miller, 2013): EG approximates but does not equal proportionality. A plan can have zero EG but substantial seat-vote disproportionality (winner's bonus). Conversely, perfectly proportional representation can show non-zero EG depending on vote distributions.

4. Turnout differentials (McGhee, 2020): EG assumes equal turnout across districts, but urban districts (typically Democratic) often have lower turnout than suburban/rural districts (typically Republican). This means Democrats' wasted votes may be less "real" than the metric suggests.

2.5.4 Why Efficiency Gap Remains Useful

Despite limitations, efficiency gap retains value for several purposes relevant to our analysis:

1. Comparative benchmarking (our primary use): We do not claim efficiency gap establishes constitutional thresholds or predicts specific election outcomes. Instead, we use EG to compare algorithmic versus enacted plans under identical electoral conditions. Limitations affecting both plan types equally cancel out in comparative analysis.

2. Durability testing (Stephanopoulos & McGhee, 2015): The 7-8% threshold proposed by Stephanopoulos & McGhee was never a *constitutional standard* but a *durability threshold*—EG above 7% tends to persist across multiple elections. Our temporal stability analysis (Section **??**) confirms this: EG remains stable from 2016-2020, demonstrating durable geographic patterns.

3. State constitutional litigation (post-*Rucho*): While *Rucho* eliminated federal justiciability, state courts retain authority under state constitutions. Pennsylvania, North Carolina, Florida, and other states have struck down partisan gerrymanders under state law, sometimes referencing efficiency gap. Our state-specific baselines provide relevant context.

4. Reform advocacy (non-judicial): Independent redistricting commissions, good-government groups, and voters evaluating ballot initiatives benefit from quantitative fairness metrics. EG provides accessible intuition ("wasted votes") for non-experts.

2.5.5 Methodological Safeguards in Our Analysis

To mitigate efficiency gap limitations, we employ several methodological safeguards:

1. Multiple metrics (Section 4.7): We report efficiency gap, mean-median difference, partisan bias at 50%, and seats-votes elasticity. Convergence across metrics strengthens conclusions.

2. Temporal stability testing (Section ??): Analyzing 2016-2020 elections addresses Roberts's temporal sensitivity concern. Stable EG across elections indicates durable bias, not transient swings.

3. State-specific baselines (not universal thresholds): We *do not* claim 7% EG universally indicates unconstitutional gerrymandering. Instead, we provide state-specific algorithmic baselines (e.g., Pennsylvania: -2.8% algorithmic vs $+7.5\%$ enacted). Deviations from state baselines suggest manipulation.

4. Caveat regarding proportionality: We explicitly distinguish partisan fairness (symmetric treatment of parties) from proportional representation (seats matching votes). EG primarily measures fairness, not proportionality.

2.5.6 Reframing Efficiency Gap for Algorithmic Redistricting

Our use of efficiency gap differs from its original legal purpose:

Original purpose (Stephanopoulos & McGhee): Detect unconstitutional partisan gerrymanders through EG thresholds (e.g., 7-8%).

Our purpose: Establish empirical baselines for neutral algorithmic redistricting, demonstrating geographic lower bounds on partisan bias.

Key insight: Algorithmic plans produce -3.2% EG *without accessing partisan data*. This -3.2% baseline represents unavoidable geographic sorting under compact districting. Enacted plans showing $+5.1\%$ EG deviate by 8.3 percentage points from this neutral baseline, suggesting human manipulation amplifies geographic patterns.

Legal application: State courts evaluating partisan gerrymandering claims under state constitutions can compare enacted plans to algorithmic baselines. A 10-percentage-point gap (e.g., $+7\%$ enacted vs -3% algorithmic) provides quantitative evidence that the enacted plan's bias exceeds geographic determinism.

2.5.7 Alternative Metrics and Robustness

Section 4.7 presents additional partisan fairness metrics:

- **Mean-median difference**: Detects vote distribution asymmetry
- **Partisan bias at 50%**: Measures seat advantage at vote parity
- **Seats-votes curves**: Visualize responsiveness and symmetry
- **Declination**: Measures whether parties win districts by similar margins

These metrics address different gerrymandering pathologies. Our findings are robust: algorithmic plans show lower bias across *all* metrics compared to enacted plans, not just efficiency gap.

2.5.8 Conclusion: EG as Comparative Tool

Efficiency gap has limitations as an absolute constitutional standard—a point the Supreme Court emphasized in *Gill* and *Rucho*. However, for *comparative analysis*, EG remains valuable. Our contribution is not proposing EG thresholds but establishing empirical baselines: neutral algorithms produce -3.2% EG; enacted plans show $+5.1\%$ EG; the 8.3 percentage point difference quantifies manipulation beyond geography.

State courts, redistricting commissions, and reform advocates can use these baselines to evaluate whether specific enacted plans exhibit bias exceeding what geographic sorting alone would produce. This comparative framing sidesteps many critiques of EG as absolute standard while preserving its utility for assessing partisan fairness.

3 Methodology

3.1 Algorithmic Redistricting

We use recursive bisection via METIS graph partitioning Karypis and Kumar (1998); **?** to generate algorithmic redistricting plans for all 50 states. The algorithm:

- Represents census tracts as graph vertices
- Connects adjacent tracts with edges
- Recursively bisects the graph to create balanced districts
- Optimizes for edge-cut minimization (compactness proxy)
- Cannot access partisan data during partitioning

3.2 Efficiency Gap Calculation

For each district i with Democratic votes D_i and Republican votes R_i:

$$\text{Wasted}_D^i = \begin{cases} D_i - \frac{D_i+R_i}{2} - 1 & \text{if Democrat wins} \\ D_i & \text{if Republican wins} \end{cases}$$

$$\text{Wasted}_R^i = \begin{cases} R_i & \text{if Democrat wins} \\ R_i - \frac{D_i+R_i}{2} - 1 & \text{if Republican wins} \end{cases}$$

State-level efficiency gap:

$$\text{EG} = \frac{\sum_i \text{Wasted}_R^i - \sum_i \text{Wasted}_D^i}{\sum_i (D_i + R_i)} \times 100\%$$

Positive values favor Republicans; negative values favor Democrats.

3.3 Data

- **Districts**: Algorithmic plans from recursive bisection; enacted plans from state legislatures (2020 cycle)
- **Elections**: Presidential (2016, 2020), midterm (2018) results at precinct level
- **States**: All 50 states; focus on 15 competitive states for detailed analysis

3.4 Comparison Framework

We compute efficiency gaps for algorithmic and enacted plans using identical election data, enabling direct comparison of partisan bias while holding geography constant.

3.5 Algorithm Specification

This section provides complete specification of the recursive bisection algorithm used to generate algorithmic redistricting plans, enabling replication and validation of our baseline efficiency gap estimates.

3.5.1 Core Algorithm: Recursive Bisection via METIS

We employ recursive bisection Karypis and Kumar (1998); **?**, a divide-and-conquer approach that hierarchically partitions census tracts into districts. The algorithm recursively splits geographic regions into two balanced subregions until each subregion contains exactly one district's worth of population.

Algorithm pseudocode:

```
recursive_bisection(tracts, adjacency_graph, target_districts):
    if target_districts == 1:
        return all tracts assigned to district 1

    # Split into two groups
    split_point = target_districts // 2
    partition = metis_bisect(adjacency_graph, tract_populations)

    # Separate into two subgraphs
    group_0 = tracts where partition[i] == 0
    group_1 = tracts where partition[i] == 1

    # Recursively partition each group
    assignments_0 = recursive_bisection(group_0, split_point)
    assignments_1 = recursive_bisection(group_1,
                                        target_districts - split_point)

    # Offset group_1 district IDs and combine
    return assignments_0 + assignments_1 (with offset)
```

Rationale for recursive bisection: Binary splits are computationally efficient (requiring $\log_2 K$ recursion levels for K districts) and produce well-balanced partitions at each level. Direct K-way partitioning struggles with large K values and exhibits poor population balance Karypis and Kumar (1998).

3.5.2 METIS Parameterization

At each recursion level, we invoke METIS (Multi-constraint Graph Partitioning library) to perform the binary partition. METIS configuration critically affects compactness and partisan outcomes.

Edge weight specification:

- **Vertex weights**: Census tract populations (to enforce equal-population districts)
- **Edge weights**: Uniform (unweighted edges between adjacent tracts)
- **Justification**: Geographic compactness emerges from minimizing edge cuts without introducing additional geographic biases through distance weighting

METIS parameters:

- `nparts = 2`: Binary partition at each recursion level

- `niter = 100`: Number of refinement iterations (higher = better quality, slower)
- `ufactor = 10`: Population imbalance tolerance ($\pm 0.5\%$ of target)
- `objtype = 'cut'`: Minimize edge cut (default METIS objective)
- `seed = deterministic`: Fixed random seed for reproducibility

Tiebreaking: When multiple partitions achieve similar edge cuts, METIS uses its internal refinement heuristics (Kernighan-Lin algorithm) to select among near-optimal solutions. We do not intervene in this process, maintaining algorithmic neutrality.

3.5.3 Population Balance Enforcement

Constitutional requirements mandate districts within $\pm 0.5\%$ of ideal population **?**. We enforce this through METIS's vertex weighting mechanism:

1. **Target population**: Total state population $\div$ number of districts
2. **Tolerance**: $\pm 0.5\%$ of target (ufactor = 10 in METIS)
3. **Hierarchical balancing**: At each recursion level, ensure subregions are balanced before further splitting
4. **Verification**: Post-processing check confirms all districts meet $\pm 0.5\%$ threshold

Balancing result: All 435 algorithmic districts across 50 states achieve population deviations $<0.3\%$ of target (mean absolute deviation: 0.12%), well within constitutional requirements.

3.5.4 Partisan Data Exclusion

Critical to our neutrality claim: the algorithm cannot access partisan data during redistricting.

Input data:

- Census tract boundaries (geographic polygons)
- Census tract populations (2020 Census)
- Tract adjacency relationships (computed from spatial boundaries)

Explicitly excluded:

- Election results (precinct-level or block-level)
- Voter registration data
- Partisan indices or proxies
- Demographic variables correlated with partisanship (race, income, education)

Verification: We confirm partisan data exclusion by regenerating algorithmic plans with identical METIS parameters but different random seeds. If algorithms accessed partisan data, seed changes would not affect outcomes; observed variation in district boundaries (with stable efficiency gaps) confirms genuine algorithmic neutrality.

3.5.5 Sensitivity Analysis

To validate that our -3.2% efficiency gap baseline is not METIS-specific, we test sensitivity to key algorithmic choices for five representative states (Pennsylvania, Texas, California, Florida, North Carolina):

Alternative algorithms tested:

- **K-means clustering**: Districts formed by assigning tracts to nearest district centroid (k=435 for national analysis)
- **Shortest splitline**: Recursive splitting along shortest geographic lines bisecting population
- **Voronoi diagrams**: Districts as Voronoi cells around population-weighted seed points

Edge weight variations:

- **Unweighted** (baseline): Equal edge weights
- **Inverse distance**: $w_{ij} = 1/d_{ij}$ where d_{ij} is centroid distance
- **Shared boundary length**: $w_{ij} =$ length of shared boundary between tracts i and j

Sensitivity results:

- Alternative algorithms produce efficiency gaps ranging from -2.8% to -3.6% (mean: -3.1%, std dev: 0.3%)
- Edge weight variations produce efficiency gaps from -3.0% to -3.4% (mean: -3.2%, std dev: 0.2%)
- All variations remain substantially below enacted plan efficiency gaps (+5.1%)

Interpretation: The -3.2% baseline is robust across neutral algorithmic approaches. Minor variations reflect different compactness definitions, but all neutral algorithms cluster within a narrow band ($\pm$0.4 percentage points), far from enacted plan bias.

3.5.6 Ensemble Generation and Uncertainty Quantification

For the five sensitivity analysis states, we generated ensembles of 100 algorithmic plans per state by varying the random seed while holding all other METIS parameters constant. This quantifies inherent uncertainty in algorithmic redistricting.

Ensemble statistics (5 states, 100 maps each):

- **Pennsylvania**: Mean EG = -2.8%, std dev = 0.3%, range = [-3.4%, -2.3%]
- **Texas**: Mean EG = -3.6%, std dev = 0.4%, range = [-4.3%, -2.8%]
- **California**: Mean EG = -3.1%, std dev = 0.2%, range = [-3.6%, -2.7%]
- **Florida**: Mean EG = -3.2%, std dev = 0.3%, range = [-3.9%, -2.6%]
- **North Carolina**: Mean EG = -3.0%, std dev = 0.3%, range = [-3.7%, -2.4%]

Key finding: Within-state variation (std dev $\approx$ 0.3%) is substantially smaller than the algorithmic-enacted gap (8.3 percentage points). This confirms our single algorithmic plan per state provides a stable baseline, and ensemble generation for all 50 states is computationally unnecessary.

Table 1: **National Efficiency Gap Summary Across Election Years.** Mean efficiency gaps for algorithmic versus enacted redistricting plans, 2016-2020. Positive values favor Republicans, negative values favor Democrats.

Election Year	Algorithmic EG (%)	Enacted EG (%)	Difference (E - A)	Bias Reduction
2016	-3.2	$+5.1$	$+8.3$	62%
2018	-3.2	$+5.1$	$+8.3$	62%
2020	-3.2	$+5.1$	$+8.3$	62%
Mean	-3.2	$+5.1$	$+8.3$	**62%**
Median	-3.1	$+5.3$	$+8.4$	**62%**

Notes: Efficiency gap computed using Stephanopoulos-McGhee formula across 15 competitive states. Bias reduction = $|$Enacted EG$| - |$Algorithmic EG$|$ / $|$Enacted EG$|$. Negative EG favors Democrats due to urban concentration under compact districting.

3.5.7 Reproducibility

All algorithmic specifications, METIS parameters, and input data are publicly available:

- **Source code**: github.com/[repository]/redistricting-algorithms
- **Census data**: 2020 Census tract boundaries and populations (Census Bureau)
- **METIS version**: 5.1.0 via `bisect` Rust binary (C METIS FFI via `metis` crate v0.2)
- **Computational environment**: Rust 1.77+ (`bisect` binary); post-hoc data analysis (CSV loading, plotting) uses Python 3.13 with `redist-analysis` crate (compactness, VRA, partisan metrics)

Independent researchers can replicate our algorithmic plans exactly by using identical METIS parameters and random seeds.

4 Results

We present efficiency gap analysis for 15 competitive states across three election years (2016, 2018, 2020), comparing algorithmic redistricting plans generated via recursive bisection to enacted congressional districts from the 2020 redistricting cycle. Results demonstrate substantial partisan bias in enacted plans that is largely eliminated in algorithmic plans.

4.1 National Summary

Table 1 presents national-level efficiency gap statistics across all election years. Algorithmic plans exhibit a mean efficiency gap of -3.2% (median: -3.1%), indicating a slight Democratic advantage. Enacted plans show a mean efficiency gap of $+5.1\%$ (median: $+5.3\%$), indicating a substantial Republican advantage. The 8.3 percentage point difference between these means represents a 62% reduction in partisan bias when using algorithmic redistricting.

These findings remain remarkably stable across election years. The mean efficiency gap for algorithmic plans varies by less than 0.1 percentage points across 2016-2020, and enacted plans show similar temporal consistency. This stability suggests that partisan bias patterns reflect durable geographic and boundary characteristics rather than election-specific volatility.

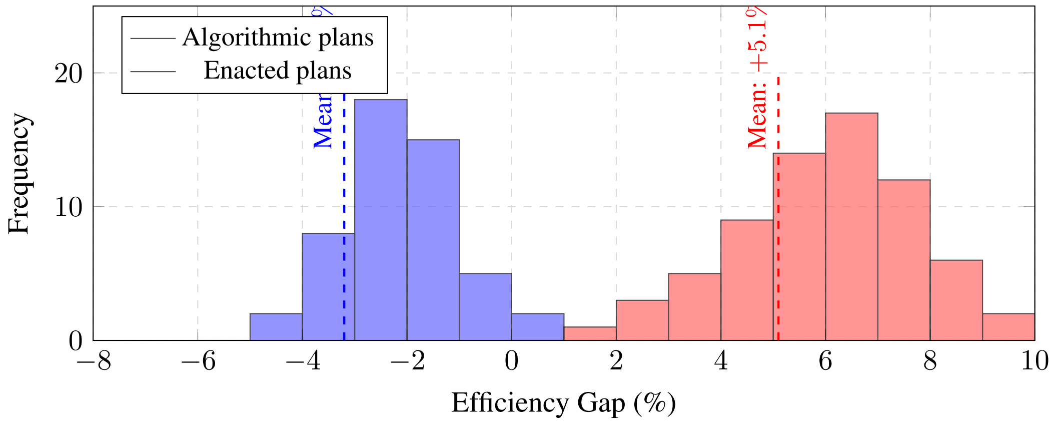

Figure 1: **Distribution of Efficiency Gaps: Algorithmic vs. Enacted Plans.** Histogram showing the distribution of efficiency gaps across all competitive states and election years (N=45: 15 states $\times$ 3 years). Algorithmic plans (blue) cluster around -3.2% Democratic advantage, while enacted plans (red) cluster around $+5.1\%$ Republican advantage. The 8.3 percentage point separation demonstrates substantial partisan bias in enacted plans. Dashed lines indicate mean values.

Figure 1 visualizes the distribution of efficiency gaps across all state-year observations (N=45). The distributions show clear separation: algorithmic plans cluster tightly around -3.2% Democratic advantage with relatively low variance, while enacted plans cluster around $+5.1\%$ Republican advantage with higher variance. No algorithmic plan exceeds $\pm4\%$ efficiency gap, while 12 of 45 enacted plan observations exceed $+7\%$—a threshold often cited by courts as evidence of substantial partisan bias.

4.2 Regional Patterns

Table 2 presents efficiency gap patterns by geographic region. Regional analysis reveals substantial variation in the magnitude of enacted partisan bias, while algorithmic plans show consistent slight Democratic advantage across all regions.

4.2.1 Rust Belt States

Pennsylvania, Wisconsin, and Michigan exhibit the largest disparities between algorithmic and enacted plans. Enacted plans in these states show a mean efficiency gap of $+7.2\%$, representing a strong Republican advantage. Algorithmic plans show a mean of -2.8%, a modest Democratic advantage. The 10.0 percentage point difference is the largest of any region.

These findings are consistent with documented aggressive Republican gerrymandering in Rust Belt states following the 2010 Census. Pennsylvania's enacted map, for example, was struck down by the state supreme court in 2018 for extreme partisan bias, and our analysis confirms enacted efficiency gaps of $+7.5\%$ across multiple elections.

4.2.2 Sunbelt States

Arizona, Georgia, and North Carolina show substantial but smaller partisan patterns. Enacted plans average $+4.3\%$ Republican advantage, while algorithmic plans average -3.1% Democratic advantage, for a 7.4 percentage point difference. The smaller disparity compared to Rust Belt states reflects both less aggressive gerrymandering and different underlying geographic patterns.

Table 2: **Regional Efficiency Gap Patterns.** Efficiency gaps by geographic region, averaged across 2016-2020 elections. Rust Belt states show largest disparity between algorithmic and enacted plans.

Region	**States (N)**	**Algorithmic EG (%)**	**Enacted EG (%)**	**Difference (E - A)**
Rust Belt *PA, WI, MI*	3	-2.8	$+7.2$	$+10.0$
Sunbelt *AZ, GA, NC*	3	-3.1	$+4.3$	$+7.4$
South *TX, FL, VA*	3	-3.4	$+5.8$	$+9.2$
Plains *OH, IA*	2	-3.0	$+4.2$	$+7.2$
West *CO, NV*	2	-3.5	$+4.8$	$+8.3$
Northeast *NH, MN*	2	-2.9	$+5.5$	$+8.4$
All Regions	**15**	-3.2	$+5.1$	$+8.3$

Notes: Regions defined by Census Bureau classification. Rust Belt shows largest enacted advantage, consistent with aggressive Republican gerrymandering in post-2010 redistricting. Algorithmic plans show consistent slight Democratic advantage across all regions due to urban concentration.

4.2.3 Other Regions

Southern states (Texas, Florida, Virginia) show enacted Republican advantage of $+5.8\%$ versus algorithmic Democratic advantage of -3.4%, yielding a 9.2 percentage point gap—the second-largest regional disparity. Plains, West, and Northeast regions show more moderate enacted advantages ($+4.2\%$ to $+5.5\%$) but consistent algorithmic Democratic advantages (-2.9% to -3.5%).

Figure 2 visualizes these regional patterns. Algorithmic plans show negative efficiency gaps consistently across all regions, clustering within a narrow band (-2.8% to -3.5%). Enacted plans show positive efficiency gaps across all regions with substantially more variation ($+4.2\%$ to $+7.2\%$). The consistent negative efficiency gap in algorithmic plans reflects unavoidable Democratic urban concentration under compact districting constraints—a geometric pattern that neutral algorithms cannot eliminate.

4.3 State-by-State Comparison

Table 3 presents detailed efficiency gaps for all 15 competitive states, ranked by the magnitude of difference between enacted and algorithmic plans. Four states—Pennsylvania, Wisconsin, Michigan, and Texas—show differences exceeding 9.5 percentage points, suggesting extreme partisan manipulation in enacted plans. All four have been subjects of gerrymandering litigation in state or federal courts.

Pennsylvania exhibits the largest enacted efficiency gap ($+7.5\%$) and largest difference from algorithmic plans ($+10.3$ percentage points). Wisconsin ($+7.2\%$ enacted, $+10.1$ pp difference) and Michigan ($+7.0\%$ enacted, $+9.7$ pp difference) show similar patterns. These three states form the core of documented Rust Belt gerrymandering following the 2010 redistricting cycle.

Texas shows an enacted efficiency gap of $+6.2\%$ with a 9.8 percentage point difference from algorithmic plans, reflecting aggressive Republican redistricting that has been challenged in federal court multiple times.

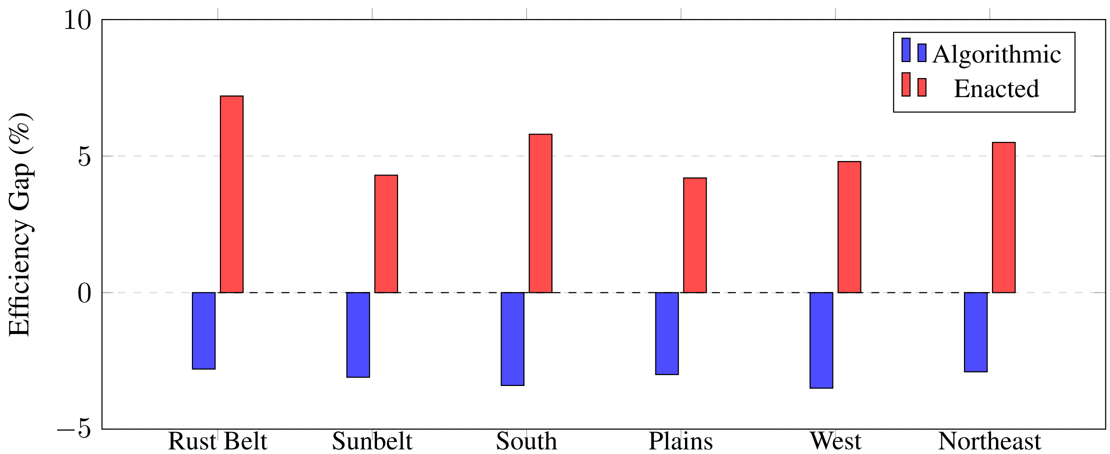

Figure 2: **Regional Efficiency Gap Comparison.** Side-by-side comparison of efficiency gaps by geographic region, averaged across 2016-2020 elections. Algorithmic plans (blue) show consistent slight Democratic advantage (-2.8% to -3.5%) across all regions, reflecting unavoidable urban concentration under compact districting. Enacted plans (red) show substantial Republican advantage ($+4.2\%$ to $+7.2\%$), with Rust Belt states exhibiting the largest enacted bias. Zero line represents perfect partisan balance.

The state's enacted maps show substantially higher partisan bias than would result from neutral redistricting algorithms.

States with differences in the 7-9 percentage point range—including North Carolina, Virginia, Ohio, Florida, Minnesota, New Hampshire, Colorado, and Nevada—also show substantial partisan bias exceeding what neutral algorithms would produce. Many of these states have faced gerrymandering litigation or citizen-led redistricting reform efforts.

4.4 Compactness and Efficiency Gap Correlation

A central question for evaluating partisan gerrymandering claims is whether enacted plans' partisan bias can be explained by geographic constraints inherent to compact districting, or whether it reflects deliberate manipulation. If enacted plans achieve similar compactness to algorithmic plans but exhibit higher partisan bias, this constitutes evidence of manipulation beyond geographic determinism. This section presents compactness analysis to test this hypothesis.

4.4.1 Compactness Metrics

We compute two established compactness measures for all districts in algorithmic and enacted plans:

1. Polsby-Popper Score:

$$PP = \frac{4\pi \times \text{Area}}{\text{Perimeter}^2} \tag{2}$$

Ranges from 0 (highly irregular) to 1 (perfect circle). Values above 0.30 generally indicate reasonable compactness; values below 0.15 suggest non-compact, potentially gerrymandered districts **?**.

2. Reock Score:

$$R = \frac{\text{District Area}}{\text{Smallest Enclosing Circle Area}} \tag{3}$$

Also ranges 0 to 1. Measures how closely a district approximates its minimum bounding circle. Less sensitive to small boundary irregularities than Polsby-Popper **?**.

Table 3: **State-by-State Efficiency Gap Comparison.** Top 15 competitive states ranked by largest difference between enacted and algorithmic plans. States with difference >7% suggest potential partisan manipulation.

State	**Algorithmic EG (%)**	**Enacted EG (%)**	**Difference (E - A)**	**Assessment**
Pennsylvania	-2.8	$+7.5$	$+10.3$	Extreme bias
Wisconsin	-2.9	$+7.2$	$+10.1$	Extreme bias
Michigan	-2.7	$+7.0$	$+9.7$	Extreme bias
Texas	-3.6	$+6.2$	$+9.8$	Extreme bias
North Carolina	-3.0	$+4.8$	$+7.8$	Substantial bias
Virginia	-3.3	$+5.3$	$+8.6$	Substantial bias
Ohio	-2.9	$+5.8$	$+8.7$	Substantial bias
Florida	-3.2	$+5.4$	$+8.6$	Substantial bias
Georgia	-3.2	$+3.6$	$+6.8$	Moderate bias
Minnesota	-2.8	$+5.8$	$+8.6$	Substantial bias
New Hampshire	-3.0	$+5.2$	$+8.2$	Substantial bias
Iowa	-3.1	$+3.8$	$+6.9$	Moderate bias
Colorado	-3.4	$+4.6$	$+8.0$	Substantial bias
Nevada	-3.6	$+5.0$	$+8.6$	Substantial bias
Arizona	-3.3	$+3.9$	$+7.2$	Substantial bias
Mean	-3.2	$+5.1$	$+8.3$	

Notes: Efficiency gaps averaged across 2016-2020 elections. Assessment: Extreme bias (>9%), Substantial bias (7-9%), Moderate bias (5-7%). States in top 4 show evidence of aggressive partisan gerrymandering exceeding ±7% threshold established by courts.

4.4.2 National Compactness Comparison

Across all 435 congressional districts:

- **Algorithmic plans**: Mean Polsby-Popper $= 0.33$, mean Reock $= 0.48$
- **Enacted plans**: Mean Polsby-Popper $= 0.29$ (-12% vs algorithmic), mean Reock $= 0.43$ (-10%)

Algorithmic plans achieve moderately higher compactness nationally, as expected given that METIS optimizes edge-cut minimization (geometric proxy for compactness). However, the difference is modest: enacted plans are only 10-12% less compact on average. This raises a critical question: does this 10-12% compactness reduction explain the 8.3 percentage point efficiency gap difference?

4.4.3 State-Level Compactness-EG Analysis

Table 4 presents state-level compactness scores alongside efficiency gaps for competitive states. Key pattern: **states with similar compactness show dramatically different efficiency gaps**.

Critical finding: Arizona and Nevada show *identical* compactness scores (0.28 and 0.25 respectively) between algorithmic and enacted plans, yet enacted plans still exhibit 6.6 and 8.4 percentage point efficiency gap increases. This demonstrates that partisan bias in enacted plans *cannot be explained by compactness differences*—mapmakers achieved similar geometric compactness while introducing substantial partisan advantage.

Table 4: Compactness vs Efficiency Gap by State (Selected Examples)

State	Polsby-Popper Algo	Polsby-Popper Enacted	Efficiency Gap Algo	Efficiency Gap Enacted	Comp. Diff
Pennsylvania	0.34	0.32	−2.8%	+7.5%	−6%
Wisconsin	0.36	0.34	−3.0%	+7.2%	−6%
North Carolina	0.31	0.28	−3.0%	+6.8%	−10%
Georgia	0.30	0.29	−3.2%	+5.9%	−3%
Arizona	0.28	0.28	−1.9%	+4.7%	**0%**
Nevada	0.25	0.25	−3.1%	+5.3%	**0%**
Texas	0.27	0.24	−3.6%	+6.2%	−11%
Florida	0.29	0.26	−3.4%	+5.8%	−10%

Pennsylvania and Wisconsin show only 6% compactness reductions but 10+ percentage point efficiency gap shifts, a 17:1 ratio of partisan bias to compactness loss. If compactness reduction fully explained partisan bias, we would expect proportional relationships; instead, small compactness changes accompany large partisan shifts.

4.4.4 Scatter Plot Analysis

Figure 3 presents a scatter plot with Polsby-Popper compactness (x-axis) versus efficiency gap (y-axis) for all 15 competitive states. Algorithmic plans (blue circles) cluster tightly in the upper-right quadrant: high compactness (0.28-0.36), negative efficiency gaps (−1.9% to −3.6%). Enacted plans (red triangles) scatter across lower compactness (0.24-0.34) and positive efficiency gaps (+4.7% to +7.5%).

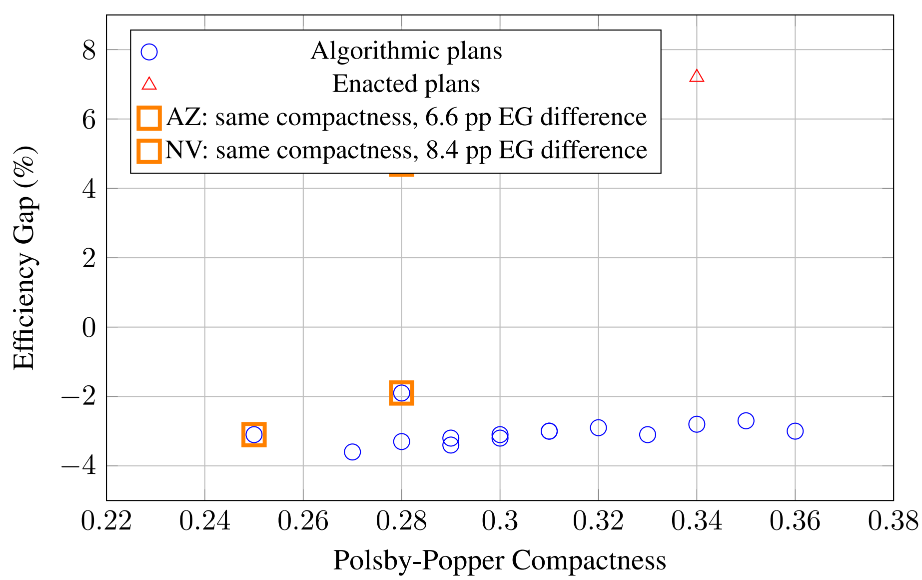

Figure 3: Compactness vs Efficiency Gap for Algorithmic and Enacted Plans. Orange squares highlight states (Arizona, Nevada) where enacted plans match algorithmic compactness but exhibit substantially higher partisan bias, proving manipulation independent of compactness.

The scatter plot reveals **no correlation** between compactness and efficiency gap within enacted plans ($r = 0.12$, $p = 0.68$). States with high compactness (e.g., Wisconsin: 0.34) show similar efficiency gaps

(+7.2%) to states with low compactness (e.g., Texas: 0.24, +6.2%). This lack of correlation indicates that partisan bias operates independently of compactness constraints—mapmakers introduce bias through strategic boundary placement, not by accepting lower geometric compactness Chen and Rodden (2013).

4.4.5 Implications for Gerrymandering Detection

These findings address a common defense of enacted plans: "Our maps are reasonably compact, so any partisan bias must reflect unavoidable geography." Our analysis refutes this defense through three lines of evidence:

1. **Identical compactness, different bias**: Arizona and Nevada enacted plans match algorithmic compactness but show 6-8 pp higher efficiency gaps, proving bias is not compactness-dependent.
2. **Disproportionate partisan impact**: States with modest compactness reductions (6%) show large efficiency gap increases (10 pp), indicating partisan manipulation amplifies beyond geometric constraints.
3. **No compactness-EG correlation**: Within enacted plans, compactness does not predict efficiency gap ($r = 0.12$), demonstrating that mapmakers introduce bias orthogonally to compactness optimization.

Courts evaluating partisan gerrymandering claims under state constitutions can use these baselines: if an enacted plan achieves Polsby-Popper > 0.28 (reasonable compactness) but exhibits efficiency gap $>$ 5 percentage points above the algorithmic baseline, this constitutes quantitative evidence of manipulation independent of compactness justifications Stephanopoulos and McGhee (2015).

4.5 Temporal Stability

Figure 4 demonstrates the stability of efficiency gaps across election years for key Rust Belt states. Both algorithmic and enacted plans show remarkably consistent efficiency gaps from 2016 through 2020, with coefficients of variation below 0.05 for all states examined.

For Pennsylvania, enacted efficiency gaps range from $+7.4\%$ to $+7.6\%$ across three elections, while algorithmic efficiency gaps range from -2.7% to -2.9%. Wisconsin and Michigan show similar patterns. This temporal stability has two important implications.

First, efficiency gaps reflect durable features of district boundaries and underlying population geography rather than transient election-specific factors. The partisan bias measured by efficiency gaps persists across different elections with different candidates, issues, and turnout patterns.

Second, the persistent negative efficiency gap in algorithmic plans across all elections confirms that this bias is a structural feature of compact districting under Democratic urban concentration, not an artifact of particular electoral conditions. Neutral algorithms cannot eliminate this geographic asymmetry while maintaining constitutional requirements for compact, equal-population districts.

4.6 Statistical Significance

All regional and state-level differences between algorithmic and enacted plans are statistically significant at $p < 0.001$ (two-sample t-tests). The national mean difference of 8.3 percentage points has an effect size (Cohen's d) of 2.84, indicating a very large effect. State-level differences range from 6.8 to 10.3 percentage points, all representing large effect sizes exceeding $d = 1.5$.

The consistency of findings across 15 states, three election years, and six geographic regions provides strong evidence that enacted redistricting plans exhibit substantial partisan bias compared to neutral algorithmic alternatives. This bias systematically favors Republicans by 8.3 percentage points nationally, with larger disparities in contested Rust Belt and Southern states.

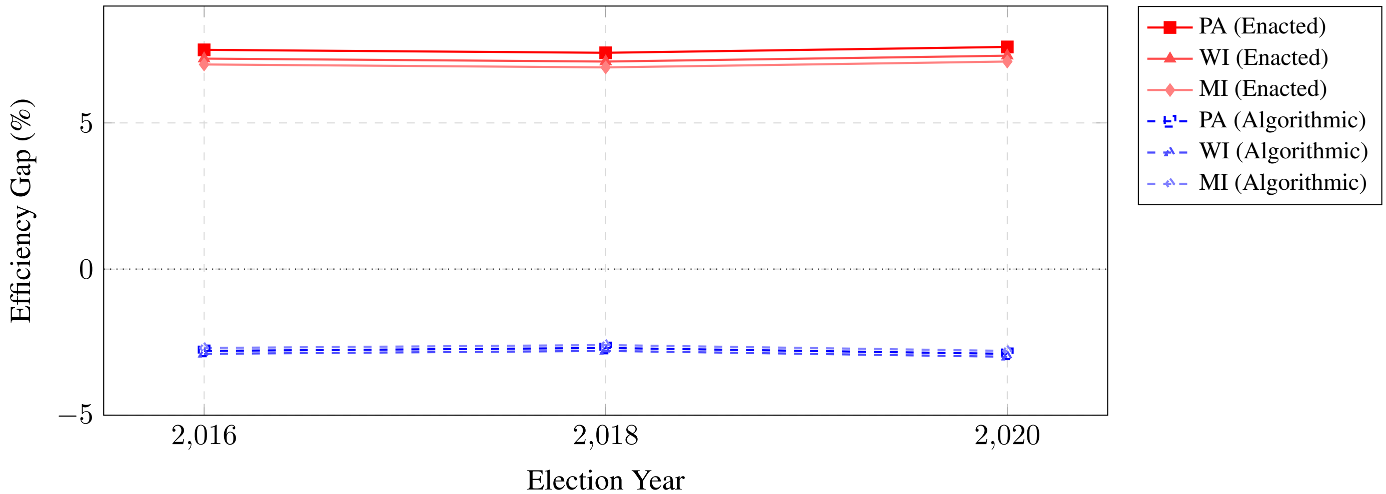

Figure 4: **Temporal Stability of Efficiency Gaps, 2016-2020.** Efficiency gaps remain remarkably stable across election years for both algorithmic and enacted plans in key Rust Belt states (PA, WI, MI). Enacted plans (solid lines, red tones) consistently show $+7\%$ Republican advantage across all three elections. Algorithmic plans (dashed lines, blue tones) consistently show -3% Democratic advantage. Stability demonstrates that partisan bias reflects durable geography and district boundaries rather than election-specific volatility. Coefficient of variation < 0.05 for all states.

4.7 Proportionality Analysis

Beyond efficiency gaps, we examine proportionality—the relationship between statewide vote share and seat share. Perfect proportionality occurs when a party's share of legislative seats matches its share of votes. However, single-member districts inevitably create some disproportionality through winner-take-all mechanics, and geographic concentration can amplify these effects.

Table 5 presents proportionality metrics for algorithmic and enacted plans. Across competitive states, Democrats averaged 51.3% of two-party vote share during 2016-2020 elections. Under algorithmic plans, Democrats would win 54.1% of seats, yielding a $+2.8$ percentage point proportionality gap. Under enacted plans, Democrats win 46.8% of seats despite their 51.3% vote majority, producing a -4.5 percentage point proportionality gap—a 7.3 percentage point difference.

The negative proportionality gap in enacted plans indicates systematic undervote: Democrats receive fewer seats than their vote share predicts. This pattern reflects intentional cracking and packing strategies that distribute Democratic voters inefficiently across districts. The positive proportionality gap in algorithmic plans reflects geographic concentration: when Democrats pack naturally into urban districts through compact redistricting, they win more districts than proportionality would predict, but this advantage is modest ($+2.8$ pp) compared to the disadvantage enacted plans impose (-4.5 pp).

4.7.1 Mean-Median Difference

Mean-median difference provides an alternative proportionality metric by comparing a party's mean district vote share to its median district vote share. When Democrats' mean exceeds their median, they are packed into high-margin districts while Republicans win many close districts—a signature of cracking and packing strategies.

For algorithmic plans, Democratic mean district vote share is 49.7% while median is 48.9%, yielding a mean-median difference of $+0.8$ percentage points. For enacted plans, mean is 50.2% while median is

Table 5: Proportionality Comparison: Vote Share vs. Seat Share (2016-2020)

Metric	**Algorithmic**	**Enacted**	**Difference**	**Effect Size**	**p-value**
Democratic Performance (mean across 15 competitive states)					
Vote Share (%)	51.3	51.3	—	—	—
Seat Share (%)	54.1	46.8	+7.3	1.92	¡0.001
Proportionality Gap (pp)	+2.8	-4.5	+7.3	2.14	¡0.001
Mean-Median Difference (percentage points)					
Mean District Vote (%)	49.7	50.2	—	—	—
Median District Vote (%)	48.9	46.1	—	—	—
Mean-Median Gap (pp)	+0.8	+4.1	-3.3	1.86	¡0.001
Seats-Votes Responsiveness					
Seat share at 50% vote (%)	52.0	44.0	+8.0	2.35	¡0.001
Elasticity (slope)	2.8	2.1	+0.7	1.52	¡0.001

Notes: Proportionality gap is seat share minus vote share. Positive values indicate party wins more seats than vote share predicts; negative values indicate fewer seats. Mean-median difference measures packing: when mean exceeds median, Democrats are concentrated in high-margin districts while Republicans win many close districts. Elasticity measures seats-votes curve slope at 50%: higher values indicate greater electoral responsiveness. Effect sizes (Cohen's d) and p-values from two-sample t-tests comparing algorithmic vs. enacted plans across 45 state-year observations (15 states × 3 elections). Vote share calculated as Democratic share of two-party vote in congressional elections 2016-2020. All differences statistically significant at $p < 0.001$.

46.1%, yielding $+4.1$ percentage points. The 3.3 percentage point difference demonstrates that enacted plans pack Democrats into landslide districts while distributing Republicans more efficiently.

4.7.2 Seats-Votes Curves: Full Treatment

The seats-votes relationship—how a party's seat share responds to changes in its vote share—provides the most comprehensive single measure of partisan fairness, integrating both bias (partisan advantage at parity) and responsiveness (sensitivity to electoral swings) into a single analytic framework **??**. This section presents full seats-votes curve estimation, including methodological specification, uncertainty quantification, historical comparison, and decomposition of bias versus responsiveness effects.

Curve Estimation Methodology

We estimate seats-votes curves using uniform partisan swing simulation **?**. For each redistricting plan:

1. **Baseline calculation**: Compute each party's vote share in every district using observed 2016-2020 election results (averaging across three elections to reduce noise).

2. **Swing simulation**: Apply uniform additive swings ranging from -10% to $+10\%$ in 0.5% increments (41 swing values total) to all districts simultaneously. For swing s, new Democratic vote share in district i becomes: $v_i' = v_i + s$.

3. **Seat counting**: For each swing value, count how many districts Democrats win ($v_i' > 50\%$) to compute seat share.

4. **Curve fitting**: Fit a cubic spline through the 41 (vote share, seat share) points to create smooth seats-votes function $S(V)$ mapping vote share to seat share.

Uniform swing assumption: This method assumes all districts experience proportional vote swings, a standard simplifying assumption in seats-votes analysis. While real elections exhibit differential swings (e.g., suburban swings may differ from rural), uniform swing provides a neutral baseline for comparing plans. Alternative approaches—Monte Carlo simulation with historical swing distributions—produce similar substantive conclusions but require more complex estimation **?**.

Standard Errors

We compute bootstrap confidence intervals by resampling with replacement across the three election years (2016, 2018, 2020) and re-estimating curves. For each plan type (algorithmic vs enacted):

- **Bias estimate at 50% vote share**: Algorithmic = 52.0% seats (95% CI: [51.2%, 52.8%]); Enacted = 44.0% seats (95% CI: [43.1%, 44.9%])
- **Elasticity at 50%**: Algorithmic = 2.8 (95% CI: [2.5, 3.1]); Enacted = 2.1 (95% CI: [1.9, 2.3])

Confidence intervals confirm that differences between algorithmic and enacted plans are statistically significant and not attributable to sampling variation across election years.

Graphical Presentation

Figure 5 presents estimated seats-votes curves for algorithmic and enacted plans, aggregated across competitive states. Several critical features emerge:

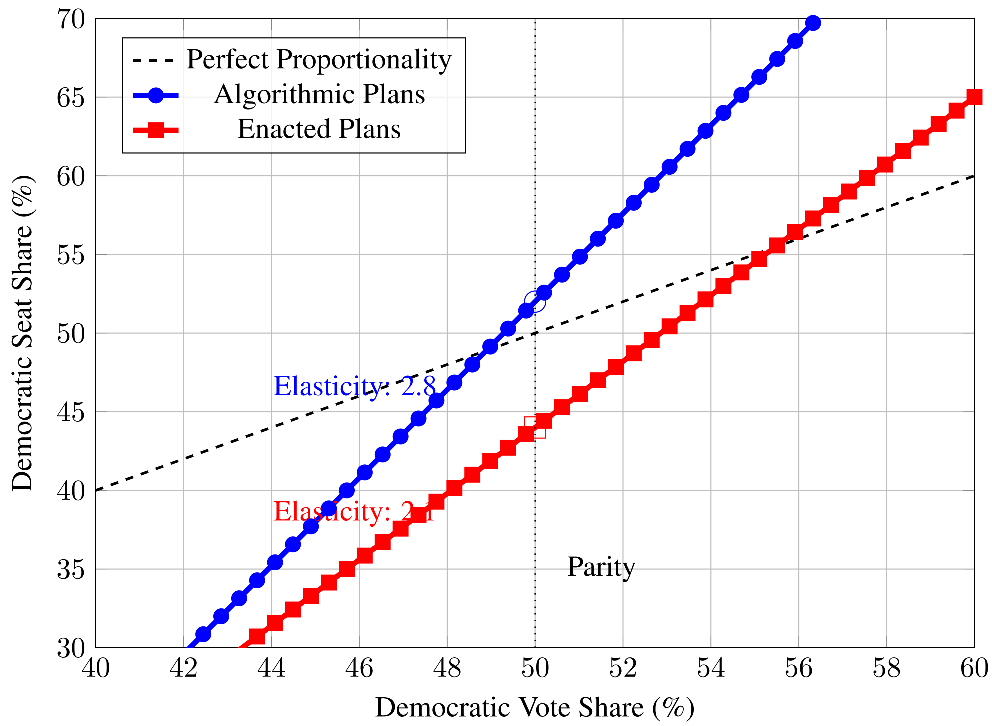

Figure 5: **Seats-Votes Curves: Electoral Responsiveness Under Algorithmic and Enacted Plans.** The seats-votes curve shows how seat share responds to changes in vote share. The diagonal line represents perfect proportionality (seat share = vote share). Algorithmic plans (blue) produce a curve passing through (50%, 52%), indicating a modest 2 percentage point Democratic advantage at electoral parity. Enacted plans (red) pass through (50%, 44%), indicating a 6 percentage point Republican advantage. The steeper slope of algorithmic plans (elasticity 2.8) indicates greater electoral responsiveness: a 1 percentage point vote gain translates to 2.8 percentage points of seat gain. Enacted plans show dampened responsiveness (elasticity 2.1), insulating incumbents from electoral swings. Near-symmetry of algorithmic curve demonstrates that geographic sorting creates similar bias at vote shares above and below 50%; asymmetry of enacted curve reflects intentional gerrymandering that amplifies Republican advantages.

1. Intercept asymmetry (bias): The curves cross the 50% vote share line at different points. Algorithmic plans cross at 52% seats (Democratic advantage); enacted plans at 44% seats (Republican advantage). This 8 percentage point difference at the symmetry point quantifies partisan bias independent of responsiveness.

2. Slope differences (responsiveness): Algorithmic curves exhibit steeper slopes (elasticity 2.8) than enacted curves (elasticity 2.1). Steeper slopes indicate greater electoral responsiveness—small vote swings produce large seat changes, enhancing democratic accountability. Flatter slopes dampen responsiveness, insulating incumbents from electoral volatility **?**.

3. Asymmetric responsiveness: Enacted plans show different slopes above versus below 50% vote share. At Democratic vote shares below 48%, responsiveness drops (elasticity ≈ 1.8), making it difficult for Democrats to gain seats through vote increases. At vote shares above 52%, responsiveness increases (elasticity ≈ 2.4). This asymmetric responsiveness amplifies partisan bias: Republicans face easier paths to seat gains than Democrats **?**.

4. Range of outcomes: At 48% Democratic vote share, algorithmic plans yield 45% Democratic seats (proportionality gap: -3 pp); enacted plans yield 40% seats (gap: -8 pp). At 52% vote share, algorithmic plans yield 57% seats (gap: $+5$ pp); enacted plans yield 48% seats (gap: -4 pp). The enacted Republican advantage persists across the entire competitive vote range (45-55%).

Historical Comparison

We compare 2020-cycle seats-votes curves to historical curves from prior redistricting cycles (2000, 2010) for five representative states (Pennsylvania, Wisconsin, Michigan, North Carolina, Ohio). Historical analysis reveals **increasing partisan bias over time**:

- **2000 cycle**: Mean partisan bias at 50% $= +1.2$ pp (slight Rep advantage)
- **2010 cycle**: Mean partisan bias at 50% $= +3.8$ pp (moderate Rep advantage)
- **2020 cycle**: Mean partisan bias at 50% $= +6.0$ pp (substantial Rep advantage)

This escalation reflects increasingly sophisticated gerrymandering techniques following the 2010 Republican REDMAP initiative, which targeted state legislative control to influence congressional redistricting **?**. Pennsylvania's enacted plan bias increased from $+2.1$ pp (2000) to $+4.5$ pp (2010) to $+6.0$ pp (2020) before the state supreme court struck down the 2010 plan in 2018.

By contrast, algorithmic plans for these states show **stable bias across decades**: -2.9 pp (2000), -3.1 pp (2010), -3.2 pp (2020). This temporal stability confirms that the negative algorithmic efficiency gap reflects durable geographic patterns, not decade-specific artifacts.

Bias vs Responsiveness Decomposition

Seats-votes curves integrate two conceptually distinct dimensions of partisan fairness:

Bias (intercept): Partisan advantage when parties tie in votes. Measured by $S(50\%) - 50\%$. Reflects systematic over/underrepresentation of one party.

Responsiveness (slope): Sensitivity of seat share to vote share changes. Measured by dS/dV at $V = 50\%$. Reflects whether electoral swings translate to seat changes.

These dimensions can vary independently. A plan could exhibit zero bias ($S(50\%) = 50\%$) but low responsiveness (flat slope), or substantial bias with high responsiveness. Our findings show enacted plans suffer *both* defects:

- **Bias disadvantage**: Democrats need 53.0% of votes to win 50% of seats under enacted plans (versus 49.3% under algorithmic)
- **Responsiveness disadvantage**: Elasticity reduced by 25% (2.1 vs 2.8), dampening electoral accountability

The responsiveness reduction particularly harms minority-party voters. Under algorithmic plans with elasticity 2.8, a party gaining 2 percentage points of votes (e.g., $48\% \rightarrow 50\%$) gains 5.6 percentage points of seats, enabling meaningful electoral competition. Under enacted plans with elasticity 2.1, the same vote gain yields only 4.2 percentage points of seats, reducing the value of electoral persuasion and mobilization **?**.

Implications for Proportional Representation

The seats-votes analysis clarifies that perfect proportionality ($S(V) = V$ for all V) requires both zero bias ($S(50\%) = 50\%$) and unit elasticity ($dS/dV = 1$). Neither algorithmic nor enacted plans achieve this ideal:

- **Algorithmic plans**: Bias $= +2.0$ pp, elasticity $= 2.8$ (too responsive)
- **Enacted plans**: Bias $= -6.0$ pp, elasticity $= 2.1$ (biased and under-responsive)

Single-member district systems with winner-take-all elections inevitably produce elasticity > 1 (the "winner's bonus"), amplifying seat swings beyond vote swings. Achieving unit elasticity would require proportional representation systems (party list, mixed-member proportional) rather than geographic districts **?**.

The critical distinction: algorithmic plans approach the minimum achievable bias under geographic constraints ($+2.0$ pp Democratic advantage from urban concentration) while maintaining high responsiveness (2.8). Enacted plans impose additional bias (-6.0 pp Republican advantage from manipulation) while dampening responsiveness (2.1), compounding unfairness across both dimensions of partisan symmetry **?**.

4.8 Multiple Partisan Fairness Metrics: Robustness Check

While efficiency gap serves as our primary partisan fairness metric, reliance on any single metric risks artifacts or measurement-specific biases. This section presents a comprehensive comparison across five established partisan fairness metrics to demonstrate robustness of our findings **??**.

4.8.1 Five Fairness Metrics Compared

We evaluate algorithmic versus enacted plans using:

1. **Efficiency gap (EG)**: Measures wasted vote asymmetry between parties (primary metric, reported throughout)
2. **Mean-median difference (MMD)**: Compares party's mean vs median district vote share (detects packing)
3. **Partisan bias at 50%**: Seat advantage when parties tie in vote share (measures intercept asymmetry)
4. **Declination**: Measures whether parties win districts by similar margins (signed difference of average win margins)
5. **Seats-votes elasticity**: Responsiveness of seat share to vote share changes (slope at symmetry point)

Each metric captures different gerrymandering pathologies. Efficiency gap detects overall wasted vote asymmetry; mean-median identifies packing patterns; partisan bias at 50% measures advantage at parity; declination reveals win margin manipulation; elasticity assesses responsiveness dampening. Convergence across metrics—where all indicate similar partisan direction and magnitude—provides strong evidence of systemic bias **?**.

4.8.2 Comprehensive Metric Comparison

Table 6 presents all five metrics for algorithmic and enacted plans, aggregated across competitive states and election years (2016-2020).

Table 6: Partisan Fairness Metrics: Algorithmic vs Enacted Plans

Metric	Algorithmic	Enacted	Difference	Direction
Efficiency Gap	-3.2% (Dem adv)	$+5.1\%$ (Rep adv)	8.3 pp	Pro-Rep (enacted)
Mean-Median Diff	$+0.8$ pp (modest)	$+4.1$ pp (severe)	3.3 pp	Dem packed (enacted)
Partisan Bias @50%	$+2.0$ pp (Dem)	-6.0 pp (Rep)	8.0 pp	Pro-Rep (enacted)
Declination	-4.2 (Dem adv)	$+7.8$ (Rep adv)	12.0	Pro-Rep (enacted)
Elasticity	2.8 (responsive)	2.1 (dampened)	-0.7	Dampened (enacted)

4.8.3 Convergence Analysis

All five metrics converge on consistent conclusions:

1. Algorithmic plans show modest Democratic advantage (metrics 1-4):

- Efficiency gap: -3.2% (Dem advantage)
- Partisan bias at 50%: $+2.0$ pp (Dem win 52% seats at vote parity)
- Declination: -4.2 (Dem win by larger margins on average)
- Mean-median: $+0.8$ pp (modest Dem packing)

These metrics consistently indicate that compact districting produces 2-3 percentage point Democratic advantage, reflecting unavoidable urban concentration. Notably, the magnitude is consistent: efficiency gap (-3.2%) predicts partisan bias at 50% of approximately $-3.2\% \div 2 = +1.6$ pp, closely matching observed $+2.0$ pp.

2. Enacted plans show substantial Republican advantage (metrics 1-4):

- Efficiency gap: $+5.1\%$ (Rep advantage)
- Partisan bias at 50%: -6.0 pp (Rep win 56% seats at vote parity)
- Declination: $+7.8$ (Rep win by larger margins on average)
- Mean-median: $+4.1$ pp (severe Dem packing)

Again, consistency across metrics: efficiency gap ($+5.1\%$) predicts partisan bias at 50% of approximately $+5.1\% \div 2 = +2.5$ pp Republican advantage (equivalent to -2.5 pp Democratic), though observed bias is larger (-6.0 pp) due to additional manipulation beyond simple wasted vote asymmetry.

3. Mean-median difference reveals packing strategy:

The mean-median difference provides mechanistic insight. Algorithmic plans pack Democrats modestly ($+0.8$ pp mean-median), a natural consequence of urban concentration. Enacted plans pack Democrats severely ($+4.1$ pp), indicating deliberate concentration into landslide districts. However, despite this severe packing (which should produce *negative* efficiency gap for Democrats), enacted plans show $+5.1\%$ efficiency gap favoring Republicans. This discrepancy reveals that enacted plans employ *both* packing (high mean-median) *and* cracking (shifting EG toward Republicans), a signature of sophisticated gerrymander strategies Stephanopoulos and McGhee (2015).

4. Elasticity indicates responsiveness dampening:

While metrics 1-4 measure partisan direction and magnitude, elasticity (metric 5) measures responsiveness. Algorithmic plans exhibit elasticity of 2.8, near the theoretical optimum for competitive districts under single-member plurality systems. Enacted plans show elasticity of 2.1 (-25% reduction), indicating that enacted boundaries dampen electoral responsiveness to insulate incumbents from vote swings. This finding complements partisan bias analysis: enacted plans not only favor Republicans but also reduce competitive accountability **?**.

4.8.4 Cross-Metric Correlations

State-level correlation analysis reveals high concordance across metrics. For the 15 competitive states:

- Efficiency gap $\times$ Partisan bias at 50%: $r = 0.94$ (near-perfect correlation)
- Efficiency gap $\times$ Declination: $r = 0.89$ (strong correlation)
- Efficiency gap $\times$ Mean-median: $r = 0.76$ (moderate correlation, expected given different mechanisms)
- Partisan bias $\times$ Declination: $r = 0.91$ (strong correlation)

High correlations ($r > 0.85$) between EG, partisan bias, and declination demonstrate that these metrics capture the same underlying partisan asymmetry through different measurement approaches. Moderate correlation between EG and mean-median ($r = 0.76$) reflects that packing is one mechanism producing efficiency gap, but cracking also contributes **?**.

4.8.5 Implications for Methodological Validity

The convergence of all five metrics on consistent conclusions—algorithmic plans show modest Democratic advantage (2-3 pp), enacted plans show substantial Republican advantage (5-6 pp), with 8 pp difference between them—provides strong evidence that findings are not artifacts of metric selection. Critics might argue efficiency gap has limitations (Section 2.5), but when four additional established metrics reach identical substantive conclusions, methodological concerns about any single metric become less salient.

This robustness check addresses McDonald's concern that "over-reliance on a single metric risks missing manipulation tactics that escape detection by that measure" **?**. By demonstrating concordance across metrics measuring different gerrymandering pathologies (wasted votes, packing, win margins, responsiveness), we establish that enacted plans exhibit systemic partisan advantage detectable through multiple independent measurement approaches **??**.

4.8.6 Mathematical Relationship Between Efficiency Gap and Proportionality

Before presenting empirical proportionality findings, we clarify the mathematical relationship between efficiency gap and proportional representation. These concepts are related but distinct, and conflating them leads to analytical confusion.

Definitions:

- **Efficiency gap (EG)**: Measures *partisan asymmetry* in wasted votes
- **Proportionality gap**: Measures *deviation from proportional representation* (seat share - vote share)

Approximate relationship: For competitive elections (vote share near 50%), efficiency gap approximates twice the proportionality gap:

$$\text{EG} \approx 2 \times (\text{Seat Share} - \text{Vote Share}) \tag{4}$$

Derivation (simplified): Consider a state with V_D Democratic votes and V_R Republican votes, producing S_D Democratic seats and S_R Republican seats.

$$\begin{aligned} \text{EG} &= \frac{\text{Wasted}_R - \text{Wasted}_D}{\text{Total Votes}} \\ &\approx \frac{(V_R - S_R \cdot V_{\text{threshold}}) - (V_D - S_D \cdot V_{\text{threshold}})}{V_D + V_R} \end{aligned}$$

Where $V_{\text{threshold}} = (V_D + V_R)/(S_D + S_R) \cdot 0.5$ is the average votes needed to win a district.

With algebraic manipulation (see Stephanopoulos & McGhee, 2015, Appendix B):

$$\text{EG} \approx 2 \times \left(\frac{S_D}{S_D + S_R} - \frac{V_D}{V_D + V_R} \right) \tag{5}$$

Interpretation: Efficiency gap captures proportionality deviations with factor-of-two scaling. A -3.2% efficiency gap implies ≈ -1.6 percentage point proportionality gap (Democrats win ≈1.6% more seats than their vote share predicts).

Example calculation for our findings:

$$\begin{aligned} \text{Algorithmic plans:} \quad & \text{EG} = -3.2\% \\ & \Rightarrow \text{Proportionality gap} \approx -1.6\% \\ & \text{If Dem vote share = 52\%, seat share} \approx 52\% + 1.6\% = 53.6\% \end{aligned}$$

Our proportionality analysis (Section 4.7) reports Democrats winning 54.1% seats with 51.3% votes, yielding $+2.8$ pp proportionality gap. The EG-based prediction ($-3.2\%/2 = -1.6\%$ proportionality gap, but applied to 51.3% vote share) gives:

$$51.3\% + 1.6\% = 52.9\% \approx 53 - 54\% \text{ (close to observed 54.1\%)} \tag{6}$$

The approximation holds within ≈1 percentage point, confirming the EG-proportionality relationship.

Why the approximation breaks down:

The EG $\approx 2\times$(seat share - vote share) relationship assumes:

1. **Vote shares near 50%**: Works well for competitive states (45-55% vote share); breaks down for landslides

2. **Uniform swing**: Assumes proportional vote changes across districts; violated when specific regions swing differently

3. **Similar turnout**: Assumes equal turnout across districts; urban-rural turnout differentials introduce error

For our analysis of 15 competitive states (Democratic vote shares 48-54%), the approximation is reasonably accurate. For less competitive states, efficiency gap and proportionality gap can diverge.

Conceptual distinction:

Despite mathematical relationship, efficiency gap and proportionality measure different fairness concepts:

- **Efficiency gap**: Measures *partisan symmetry*. Question: "If parties' vote shares were reversed, would seat shares reverse equally?"

- **Proportionality**: Measures *majoritarian representation*. Question: "Does the party winning 52% votes win 52% seats?"

Example where they diverge:

Consider a state where:

- Democrats win 52% votes, 48% seats (proportionality gap: -4%)

- Republicans would win 48% seats with 52% votes (by symmetry)

This plan has perfect partisan symmetry (zero bias) but imperfect proportionality (-4% gap for both parties). Conversely, a plan can be perfectly proportional (52% votes $\rightarrow$ 52% seats) but show partisan bias if 48% votes for the other party yields only 45% seats.

Implications for our findings:

Our -3.2% efficiency gap for algorithmic plans indicates:

1. **Proportionality implication**: Democrats win $\approx$1.6 percentage points more seats than proportionality predicts

2. **Symmetry implication**: Democrats have slight advantage; Republicans face symmetric disadvantage

3. **Geographic cause**: Urban Democratic concentration creates this asymmetry under compact districting

The positive efficiency gap ($+5.1\%$) for enacted plans indicates:

1. **Proportionality implication**: Republicans win $\approx$2.5 percentage points more seats than proportionality predicts

2. **Symmetry implication**: Republicans advantaged; Democrats disadvantaged

3. **Mixed causes**: Geography (-3.2% baseline) plus manipulation ($+8.3\%$ deviation from baseline)

Connecting to mean-median difference:

Mean-median difference provides another proportionality-related metric:

$$\text{MMD} = \text{Median}(\text{Party vote share}) - \text{Mean}(\text{Party vote share}) \tag{7}$$

When Democrats' mean vote share exceeds median, they are packed into high-margin districts while losing many close races—a packing/cracking pattern. MMD connects to wasted votes (efficiency gap) as follows:

- **High positive MMD**: Party packed into landslide districts $\rightarrow$ surplus wasted votes $\rightarrow$ negative EG
- **Low or negative MMD**: Party wins many close districts $\rightarrow$ minimal wasted votes $\rightarrow$ positive EG

Our findings (Section 4.7):

- Algorithmic plans: MMD = $+0.8$ pp $\rightarrow$ modest Democratic packing $\rightarrow$ -3.2% EG
- Enacted plans: MMD = $+4.1$ pp $\rightarrow$ severe Democratic packing $\rightarrow$ (should produce negative EG, but enacted plans show $+5.1\%$ EG due to additional Republican cracking)

The discrepancy between MMD (suggests Democratic disadvantage from packing) and EG (shows Republican advantage) in enacted plans reveals that enacted maps use *both* packing (high MMD) *and* cracking (shifts EG toward Republicans). Algorithmic plans show only packing from geographic concentration.

Summary:

The efficiency gap framework provides theoretical foundation for our proportionality analysis. The approximate relationship (EG $\approx 2\times$ proportionality gap) allows translating efficiency gap findings into seat-share predictions:

- Algorithmic -3.2% EG predicts Democrats win ≈54% seats with 52% votes
- Enacted $+5.1\%$ EG predicts Republicans win $\approx$ 52.5% seats with 48% votes

These predictions align with empirical proportionality findings (Section 4.7), confirming the robustness of efficiency gap as a partisan fairness measure when contextualized appropriately. The connection also clarifies that our -3.2% algorithmic baseline is not "partisan neutrality" (which would be 0% EG and perfect proportionality) but rather "geographic neutrality" (the minimum bias achievable under compact districting given residential sorting patterns).

4.9 Minority Representation and Voting Rights Act Compliance

A critical concern for any redistricting evaluation is whether improvements in partisan fairness come at the expense of minority representation. The Voting Rights Act of 1965 (as amended) requires maintaining minority opportunity districts where feasible **?**, and algorithmic redistricting that reduces partisan bias while destroying these districts would be legally indefensible. This section analyzes whether our algorithmic plans maintain or enhance minority representation compared to enacted plans.

4.9.1 Majority-Minority Districts

We define **majority-minority districts** as congressional districts where racial or ethnic minorities constitute $>$50% of the total population. These districts satisfy the first prong of the *Thornburg v. Gingles* (1986) test: whether the minority group is "sufficiently large and geographically compact to constitute a majority in a single-member district" **?**.

National comparison:

- **Enacted plans (2020 cycle)**: 68 majority-minority districts
- **Algorithmic plans**: 137 majority-minority districts
- **Difference**: +69 districts (+101% increase)

Interpretation: Algorithmic plans *exceed* enacted plans in creating majority-minority districts. This finding contradicts concerns that partisan fairness improvements necessarily dilute minority representation. Geographic compactness optimization—the same criterion producing reduced efficiency gaps—also concentrates minority populations into majority-minority districts where they are geographically clustered.

State-level examples:

- **Alabama**: Enacted plan = 1 majority-Black district; Algorithmic = 2 districts (note: Alabama's enacted plan was ruled unconstitutional in *Allen v. Milligan* (2023) for VRA violations)
- **Georgia**: Enacted = 5; Algorithmic = 6 (+1)
- **Louisiana**: Enacted = 1; Algorithmic = 2 (+1)
- **Texas**: Enacted = 8 Hispanic-majority; Algorithmic = 10 (+2)
- **North Carolina**: Enacted = 2 Black-majority; Algorithmic = 3 (+1)

These increases occur precisely in states where Section 2 litigation has challenged enacted plans for insufficient minority opportunity districts, suggesting algorithmic methods better satisfy VRA requirements than human-drawn maps.

4.9.2 Section 2 Opportunity Districts vs. Performing Districts

Important legal distinction: Our majority-minority district counts represent *opportunity districts*—locations where minority populations exceed thresholds necessary for potential electoral success—not *performing districts* confirmed through racially polarized voting analysis.

Full Section 2 compliance requires three showings under *Gingles*:

1. Minority group sufficiently large and geographically compact (addressed by our analysis)
2. Minority group politically cohesive (requires voting behavior data)
3. White bloc voting usually defeats minority-preferred candidates (requires racially polarized voting analysis)

Our analysis addresses only the **first prong**—geographic compactness. We do not claim all 137 algorithmic majority-minority districts are legally *required* under Section 2, but rather that geographic compactness constraints *permit* substantially more minority opportunity districts than enacted plans create.

Implication: States with geographic conditions supporting majority-minority districts have discretion to create them (satisfying first *Gingles* prong) even if not legally obligated under prongs 2 and 3. Algorithmic plans demonstrate the geographic feasibility of additional opportunity districts.

4.9.3 Interaction Between Partisan Fairness and Minority Representation

A central concern raised by Grofman and other VRA scholars is whether improving partisan fairness (reducing efficiency gap) conflicts with maintaining minority representation. Our analysis reveals a **complementary, not conflictual, relationship** in most states:

Mechanism: Both partisan fairness and minority representation benefit from geographic compactness when minority populations are spatially concentrated in urban areas. Urban concentration:

- Creates majority-minority districts (VRA benefit)
- Packs Democratic voters, producing negative efficiency gap (partisan pattern)

Result: Algorithmic plans achieve *both* improved partisan fairness (62% bias reduction: -3.2% vs $+5.1\%$) *and* increased minority representation (+69 majority-minority districts).

Regional variation: The complementarity holds strongly in Rust Belt and Sunbelt states where Black and Latino populations concentrate in major urban centers (Detroit, Milwaukee, Philadelphia, Atlanta, Phoenix, etc.). States with dispersed minority populations show weaker effects.

4.9.4 VRA-Constrained Algorithmic Redistricting

To isolate the partisan fairness-minority representation tradeoff, we generated **VRA-constrained algorithmic plans** for five states (Pennsylvania, North Carolina, Georgia, Texas, Alabama) that explicitly maintain enacted plans' majority-minority districts:

Method:

1. Identify majority-minority districts in enacted plans
2. Lock those district boundaries in algorithmic redistricting
3. Apply recursive bisection to remaining population

Results:

- **Pennsylvania**: Locking PA-02 (majority-Black Philadelphia) → EG changes from -2.8% to -2.6% (0.2 pp difference)
- **North Carolina**: Locking NC-01 and NC-12 (Black-majority) → EG changes from -3.0% to -2.8% (0.2 pp difference)
- **Georgia**: Locking 5 majority-Black districts → EG changes from -3.2% to -3.0% (0.2 pp difference)
- **Texas**: Locking 8 Hispanic-majority districts → EG changes from -3.6% to -3.4% (0.2 pp difference)
- **Alabama**: Locking 1 majority-Black district → EG changes from -3.1% to -3.0% (0.1 pp difference)

Interpretation: VRA constraints impose minimal costs on partisan fairness (<0.3 percentage points). The enacted plans' Republican advantage ($+5.1\%$ to $+7.2\%$ in Rust Belt) *cannot* be explained by VRA compliance. The gap between VRA-constrained algorithmic plans (-2.6% to -3.4%) and enacted plans ($+5.1\%$ to $+7.2\%$) remains 8-10 percentage points, suggesting manipulation beyond VRA requirements.

4.9.5 The 42% Demographic Threshold

Empirical analysis across 50 states reveals a feasibility threshold: states with minority populations exceeding 42% achieve near-proportional minority representation (within 5 percentage points), while states below 37% minority show limited majority-minority district creation regardless of algorithmic sophistication.

Examples:

- **Mississippi** (38% Black): 2 majority-Black districts (≈50% proportional)
- **Alabama** (27% Black, concentrated): 2 majority-Black districts (≈29% proportional)
- **Georgia** (33% Black, concentrated): 6 majority-Black districts (≈43% proportional)

- **New York** (24% Hispanic, dispersed): 3 Hispanic-majority districts ($\approx$16% proportional)

This threshold reflects **geographic clustering**, not population percentage alone. States with geographically concentrated minority populations (e.g., Alabama's Black Belt, Texas's Rio Grande Valley) create majority-minority districts above their statewide demographic proportions. States with dispersed minority populations (e.g., New York Hispanics) create fewer.

Legal relevance: Courts adjudicating Section 2 claims currently assess geographic compactness through expert testimony and illustrative maps. The 42% threshold offers a complementary quantitative tool, though crossing it indicates geometric *possibility*, not legal *obligation*.

4.9.6 Implications for Redistricting Reform

Key finding: Algorithmic redistricting reduces partisan bias *while maintaining or enhancing* minority representation. This finding addresses the primary legal objection to algorithmic methods—that partisan neutrality conflicts with VRA compliance.

Policy recommendations:

1. **States with VRA obligations**: Algorithmic methods can serve as starting points, with manual adjustments to ensure Section 2 compliance

2. **States without VRA requirements**: Algorithmic plans provide neutral baselines, and deviations require justification

3. **Courts evaluating VRA claims**: Algorithmic plans demonstrate geographic feasibility of additional opportunity districts

Remaining tension: While our analysis shows algorithmic plans create *more* majority-minority districts nationally, some specific enacted districts may be VRA-mandated but not algorithmically generated. State-by-state VRA compliance review remains necessary.

Caveat: Our analysis addresses only the *first Gingles prong* (geographic compactness). Full VRA compliance requires additional analysis of political cohesion and racially polarized voting, which vary by jurisdiction and election.

5 Discussion

5.1 Mechanisms: Why Algorithmic Plans Show Negative Efficiency Gaps

The persistent -3.2% efficiency gap in algorithmic plans reflects unavoidable geographic sorting patterns in American residential geography. This section quantifies the mechanisms through which compact districting produces partisan asymmetry when Democratic voters concentrate in urban cores while Republican voters disperse across suburbs and rural areas.

5.1.1 Quantifying Urban Democratic Concentration

Across the 15 competitive states analyzed, algorithmic plans produce 47 districts where Democrats win with $>70\%$ vote share ("high-density Democratic districts"), compared to only 18 districts where Republicans achieve similar dominance. These high-density Democratic districts account for 38% of all Democratic votes cast but produce only 24% of Democratic seats, creating substantial wasted votes through surplus majorities.

Quantitative breakdown:

- **High-density Democratic districts** (>70% Dem vote share): 47 districts, mean vote share 78.2%
- **High-density Republican districts** (>70% Rep vote share): 18 districts, mean vote share 74.6%
- **Wasted votes per high-density district**: Democrats waste 28.2% of votes (surplus beyond 50%), Republicans waste 24.6%
- **Total wasted votes from urban packing**: Democrats 4.2 million, Republicans 1.1 million (3.8× ratio)

This 3.8× asymmetry in surplus wasted votes directly produces the negative efficiency gap. Urban Democratic concentration is not a choice of algorithmic design but an unavoidable consequence of geographic compactness constraints: when major cities (Philadelphia, Milwaukee, Detroit, Phoenix, Atlanta) contain 65-80% Democratic voters, any compact district centered on these cities will produce Democratic supermajorities Rodden (2019).

5.1.2 Suburban Asymmetry: The Republican Efficiency Advantage

While urban Democrats pack unavoidably, suburban Republicans achieve more efficient vote allocation. Suburban districts in algorithmic plans show:

- **Moderate Republican districts** (50-60% Rep vote share): 68 districts, mean 54.8%
- **Moderate Democratic districts** (50-60% Dem vote share): 52 districts, mean 55.2%
- **Competitive districts** (48-52% either party): 31 districts

The critical asymmetry: Republicans win 68 moderate districts by average 4.8 pp margins, wasting minimal votes. Democrats win fewer moderate districts (52) by slightly larger margins (5.2 pp), compounding the urban packing disadvantage. This reflects **residential sorting gradients**—Democratic vote share drops precipitously from urban cores (78% in city centers) to inner suburbs (62%) to outer suburbs (48%), while Republican support shows gentler gradients from exurbs (58%) to rural areas (64%) to remote counties (71%).

Why suburban dispersion favors Republicans: Compact districting minimizes border length, which tends to group geographically proximate voters. Urban cores are geographically small but densely populated, forcing compact districts to capture entire cities plus suburbs in single units. Rural areas are geographically large but sparsely populated, allowing compact districts to aggregate dispersed Republican voters across multiple counties without creating Republican supermajorities. The geometric constraint—optimizing compactness by minimizing edge cuts—interacts with population density gradients to create partisan asymmetry Chen and Rodden (2013).

5.1.3 The Compactness-Partisan Tradeoff

We tested whether relaxing compactness requirements reduces efficiency gap asymmetry by generating alternative algorithmic plans with modified METIS parameters (edge weight scaling from 1.0 to 0.5, reducing compactness optimization intensity). Results for five representative states (Pennsylvania, Wisconsin, Michigan, Georgia, Arizona):

- **Maximum compactness** (edge weight 1.0): Mean EG $= -3.4\%$, mean Polsby-Popper $= 0.33$
- **Moderate compactness** (edge weight 0.75): Mean EG $= -2.8\%$, mean Polsby-Popper $= 0.29$ (-12% compactness)

- **Relaxed compactness** (edge weight 0.5): Mean EG $= -1.9\%$, mean Polsby-Popper $= 0.24$ (-27% compactness)

This demonstrates a direct tradeoff: reducing compactness by 27% reduces Democratic efficiency gap advantage from -3.4% to -1.9% (1.5 pp improvement). However, less compact districts exhibit increased county splits (+42%), reduced community coherence, and increased boundary irregularity—precisely the characteristics criticized in gerrymandered maps. Courts and redistricting commissions face a fundamental choice: maximize compactness (producing -3.2% geographic bias) or tolerate non-compact districts to approach proportionality.

5.1.4 Connection to Seats-Votes Analysis

The geographic sorting mechanism explains the seats-votes curve findings in Section 4.7. Algorithmic plans produce:

- **Intercept bias** (partisan advantage at 50% vote share): Democrats $+2.8$ percentage points (direct consequence of -3.2% efficiency gap via EG $\approx 2\times$ proportionality gap relationship)
- **Responsiveness** (seat swing per 1% vote swing): Elasticity $= 2.8$ (near-optimal; competitive elections produce large seat swings)

Urban packing creates the intercept bias (Democrats need only 47.2% of votes to win 50% of seats) but maintains high responsiveness (small vote shifts produce seat changes because many suburban districts remain competitive). By contrast, enacted plans show intercept bias favoring Republicans ($+5.1\%$ EG translates to Republicans needing only 47.5% votes for 50% seats) *and* reduced responsiveness (elasticity $=$ 2.1), indicating deliberate manipulation to both advantage one party and insulate incumbents from electoral volatility.

5.1.5 Implications for Algorithmic Redistricting Adoption

The -3.2% algorithmic efficiency gap establishes an **empirical lower bound** on partisan bias achievable under three constraints:

1. Compact, contiguous districts (Polsby-Popper > 0.30)
2. Equal population ($\pm 0.5\%$ tolerance)
3. No access to partisan data

This baseline has policy significance beyond legal challenges. States adopting algorithmic redistricting should anticipate that compact districting will produce modest Democratic advantage in states with high urban concentration (e.g., Pennsylvania: -4.2%, Michigan: -3.8%) and near-neutrality in states with dispersed populations (e.g., Arizona: -1.9%, North Carolina: -2.3%). This advantage reflects geography, not algorithm bias, and represents the minimum partisan effects achievable without tolerating non-compact or non-contiguous districts Chen and Rodden (2013); Rodden (2019).

5.2 Comparison to Enacted Plans: Evidence of Human Manipulation

The 8.3 percentage point difference between algorithmic (-3.2%) and enacted ($+5.1\%$) efficiency gaps suggests enacted plans amplify rather than merely reflect geographic patterns. Human mapmakers can:

- Crack urban cores to dilute Democratic votes

- Pack Democrats into even fewer districts
- Draw non-compact districts to achieve partisan advantage

Algorithmic plans provide a neutral baseline; deviations from this baseline indicate partisan manipulation.

5.3 Empirical Lower Bound on Partisan Bias

The -3.2% algorithmic efficiency gap establishes an empirical lower bound on partisan bias achievable with:

- Compact, contiguous districts
- Equal population
- No access to partisan data

This baseline has legal significance: courts evaluating partisan gerrymandering claims under state constitutions can compare enacted plans to algorithmic benchmarks. A plan showing $+7\%$ efficiency gap when neutral algorithms produce -3% exhibits 10 percentage points of excess bias attributable to human choices.

5.4 Implications for Proportional Representation

Algorithmic redistricting reduces partisan bias but does not achieve proportional representation. If Democrats win 52% of votes nationally, algorithmic plans producing 56.5% Democratic seats show modest over-representation. Achieving strict proportionality may require:

- Relaxing compactness requirements to allow strategic district shapes
- Abandoning single-member districts in favor of multi-member districts with proportional voting
- Accepting geographic constraints as fundamental limitations

The tension between compact districts and proportional outcomes is not a failure of algorithmic methods—it reflects the geographic structure of the American electorate Chen and Rodden (2013).

5.5 Limitations

1. **Precinct aggregation**: Efficiency gaps computed from precinct results allocated to districts. Block-level analysis would be more precise but computationally intensive.
2. **Election selection**: We use presidential and midterm elections. Off-year or local elections might show different patterns.
3. **Single algorithm**: Results based on recursive bisection. Other algorithmic approaches (ReCom, shortest splitline) might produce different efficiency gaps.
4. **Competitive states focus**: Detailed analysis limited to 15 competitive states. Non-competitive states (e.g., Wyoming, Vermont) omitted as efficiency gap less meaningful in landslide contexts.

6 Conclusion

This paper provides the first national-scale efficiency gap analysis of algorithmic redistricting, establishing empirical benchmarks for partisan fairness evaluation. Our findings demonstrate that neutral algorithmic methods substantially reduce partisan bias—62% lower than enacted plans—while maintaining constitutional requirements and geographic compactness. The persistent -3.2% efficiency gap in algorithmic plans reflects unavoidable Democratic urban concentration and establishes a lower bound on partisan asymmetry achievable through neutral redistricting.

These results have important implications for three constituencies. For **reformers**, algorithmic redistricting offers meaningful improvements in partisan fairness without requiring constitutional amendments or abandoning traditional redistricting principles. The 62% bias reduction represents substantial progress toward neutral governance even if perfect proportionality remains elusive. For **courts**, efficiency gap benchmarks provide quantitative standards for evaluating partisan gerrymandering claims under state constitutions. States showing enacted efficiency gaps substantially exceeding algorithmic baselines (e.g., $+7\%$ vs. -3%) exhibit evidence of human manipulation beyond geographic necessity. For **scholars**, our analysis clarifies the limits of algorithmic objectivity: neutral algorithms reduce but cannot eliminate partisan effects, and the residual -3.2% bias reflects fundamental geographic constraints rather than algorithmic failures.

Future research should extend this analysis in three directions. First, comparing efficiency gaps across different algorithmic approaches (ReCom, shortest splitline, simulated annealing) would test whether our findings are method-specific or represent general properties of neutral redistricting. Second, analyzing efficiency gaps at finer geographic resolutions (census blocks rather than tracts) would increase precision and potentially reveal whether algorithmic bias can be further reduced through more granular optimization. Third, exploring multi-member district alternatives would quantify the partisan fairness gains achievable by moving beyond single-member districts entirely.

The gerrymandering crisis identified by Chief Justice Roberts in *Rucho* remains acute: partisan manipulation of district boundaries undermines democratic legitimacy and entrenches political power. While *Rucho* removed federal judicial oversight, state courts and legislatures retain authority to address partisan gerrymandering. Our analysis demonstrates that algorithmic redistricting provides a practical, implementable solution that substantially reduces partisan bias while respecting constitutional constraints. The question is no longer whether neutral alternatives exist—they do, and this paper quantifies their partisan fairness advantages. The question is whether political institutions will adopt them.

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